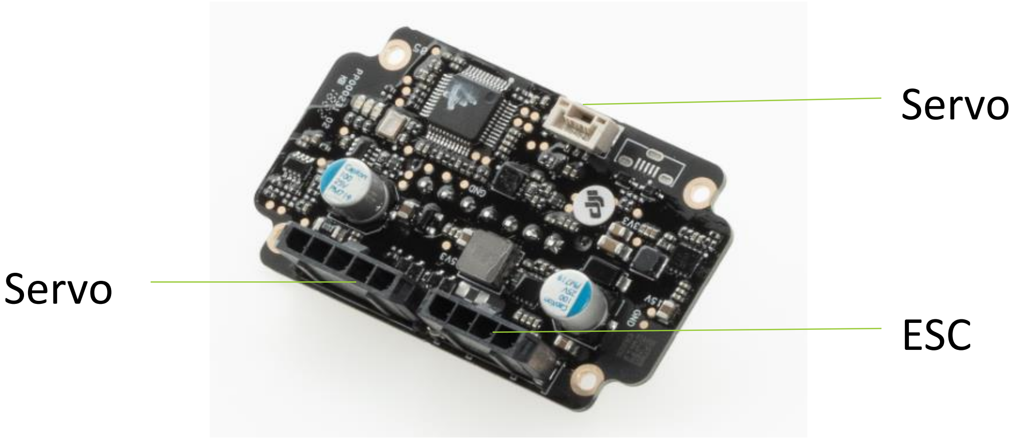
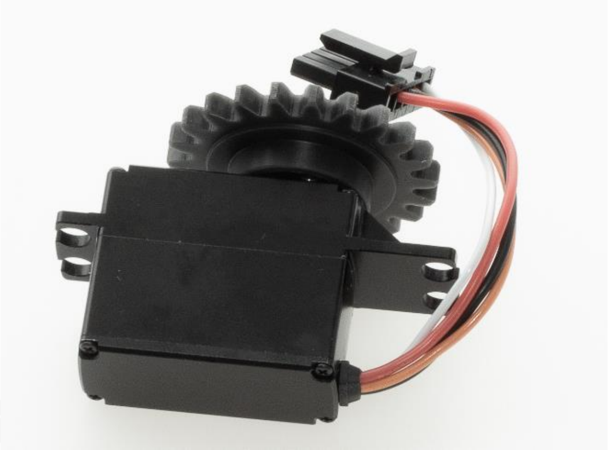
Spreading System Repair Training

Contents

01 Module Principle





- Control board

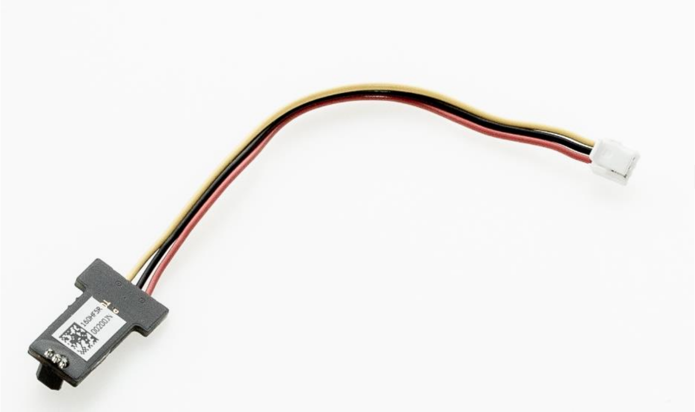
- Servo assembly

Control board: The Spreading System main board receives the aircraft's control signal and provides feedback information about the spreading system.
Servo: Controls the Spreading System hopper outlet size.

- Reduction gearbox
- ESC module

Reduction gearbox: Contains a motor and control the Spreading System rotation.

ESC module: Receives the control board information to control the reduction gearbox.

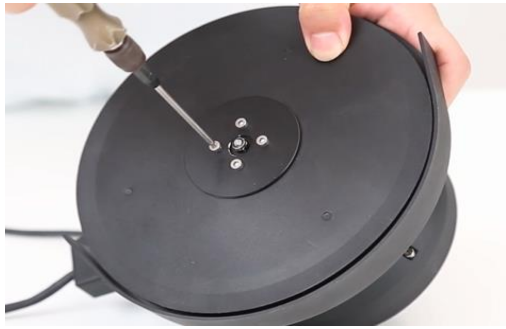


- Hall sensor
- Hall sensor board

The Hall sensor is mounted on the reduction gearbox. The Hall sensor board detects the rotation frequency of the Hall sensor and calculates the rotation speed of the spreading system, and then provides corresponding information to the control board.

If the app prompts that the tank is empty but there is remaining liquid in the tabk, please check the Hall sensor and the Hall sensor board first.

02 Assembly & Disassembly



1) Remove the spinner disk's washer.

2) Remove the anti-skid screws with the sleeve, then remove the spinner disk.

3) Loosen the set screw, then remove the flange.

4) Remove the Funnel.

5) Remove the Funnel.

6) Rotate the small gear counterclockwise to its limit.

7) Remove the screw securing the servo disk.

8) Remove the four screws securing the small gear.

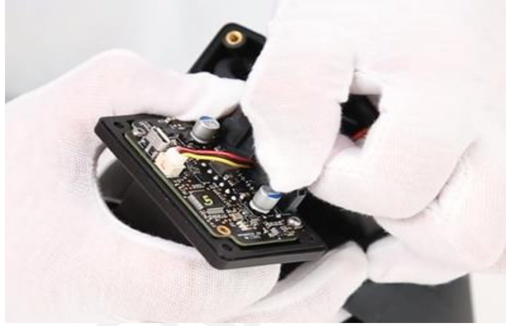
9) Remove the large gear.

10) Remove the small gear.

11) Remove the servo disk.

12) Remove the screw securing the Spreading System lock knob, then remove the Spreading System lock knob.

13) Remove the outer cover of the control box.

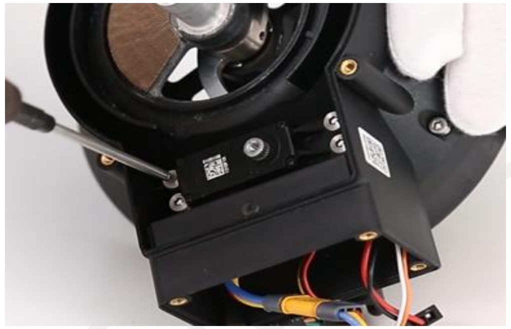


14) Disconnect all the cables on the control board.

15) Remove the screws securing the control board.

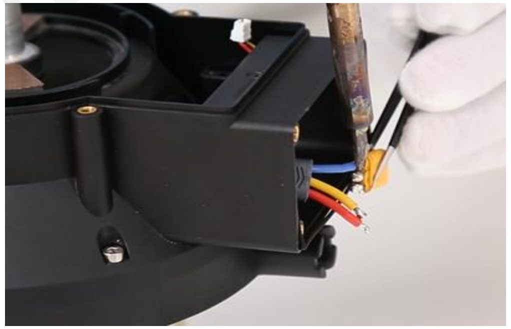
16) Push out the control cable.

17) Separate the control board and control cable.



18) Remove the servo.

19) Remove the ESC board.

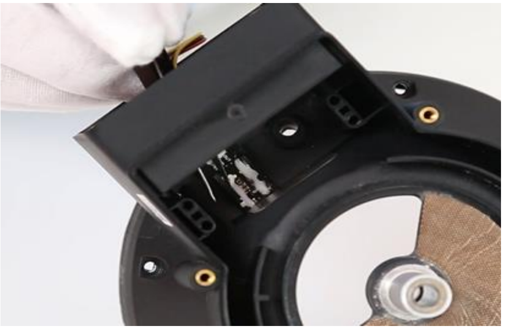


20) Desolder the ESC cable connector with the electronic soldering iron.

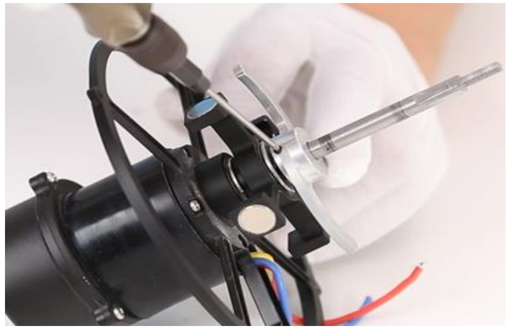
21) Remove the circlip at the arrow. Remove the four screws securing the spinner disk and reduction gearbox.

22) Disconnect the ESC cable.

23) Separate the reduction gearbox and spinner disk.

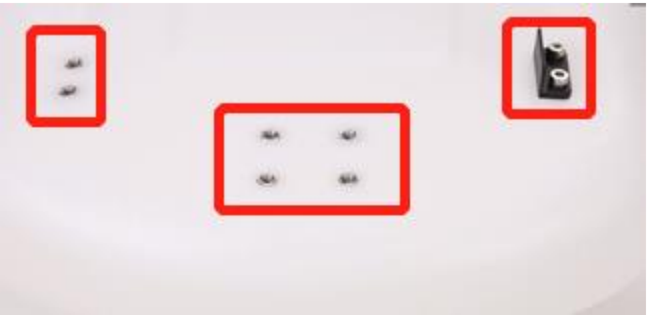


24) Clean off the white glue on the Hall sensor board and remove the Hall sensor board.

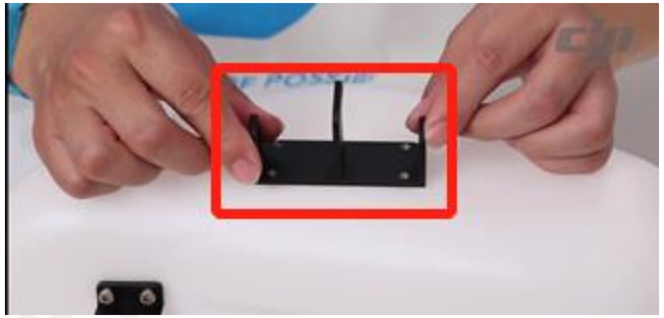


25) Loosen the set screw on the muddler, then remove the muddler.

26) Remove the long screws securing the spread tank's cover, then remove the spread tank's cover.

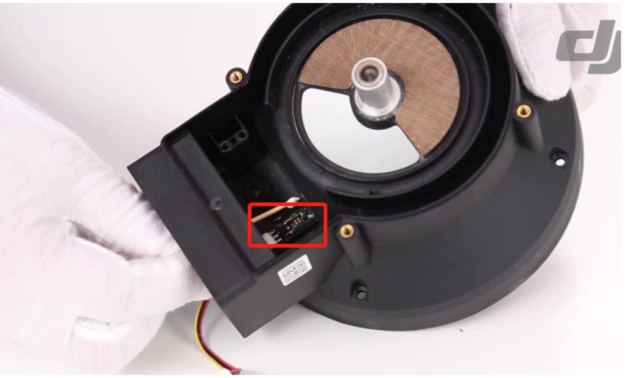


1) Mount battery limit A and two battery limit Bs, and secure them with eight screws.



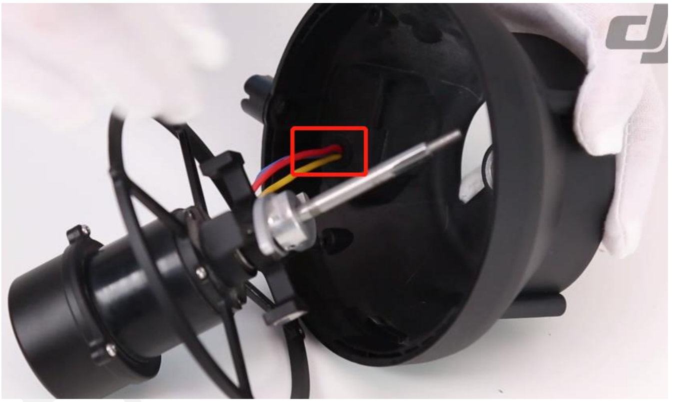
2) Mount the spread tank support, and secure it with four screws.

3) Mount the spread tank's cover, and secure it with one screw.



4) Mount the Hall magnet into the spinner disk, and apply an appropriate amount of white glue.

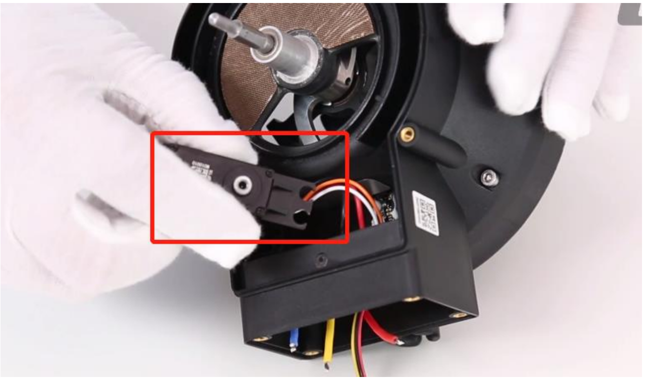
5) Mount the muddler, and secure it with two screws (screw glue should be applied).



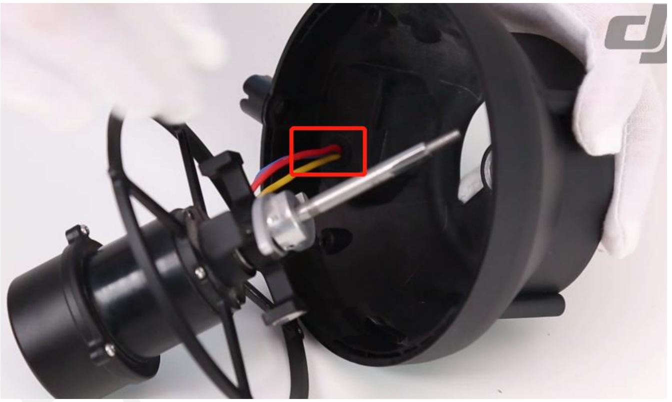
6) Pass the motor cable through the spinner disk.

7) Pass the motor shaft into the spinner disk.

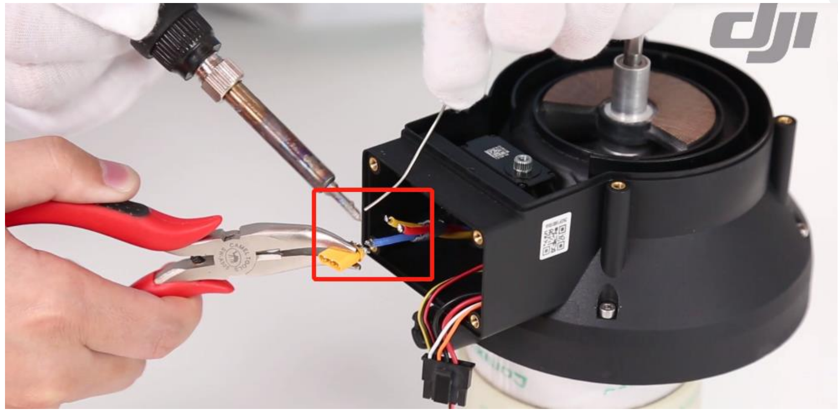
8) Adjust the positions of the reduction gearbox and spinner disk. Align with the screw holes. Secure them with four screws (screw glue should be applied).



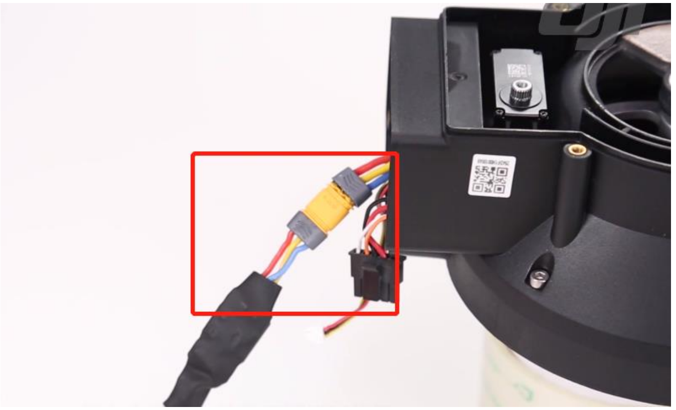
9) Mount the servo and secure it with four screws (the servo cable is at the Hall sensor board side).



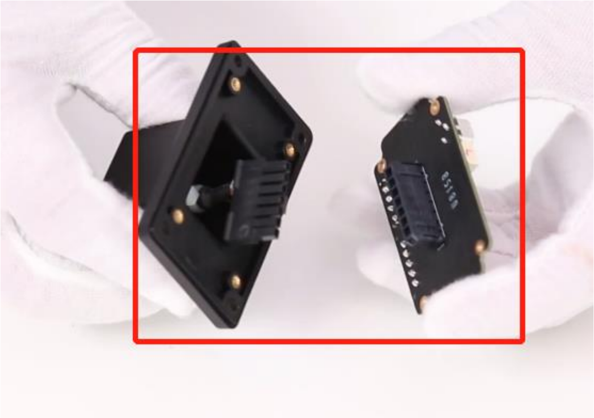
10) Pass the motor cable through the spinner disk.



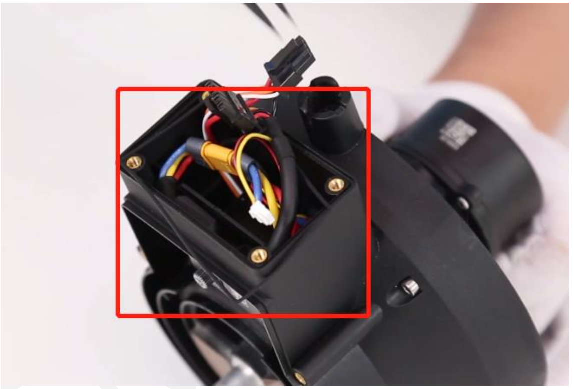
11) Weld the motor cables on the motor cable connector in order. (The red, yellow and blue cables correspond to Connectors 1, 2, and 3)



12) Connect the ESC board, then arrange the cables



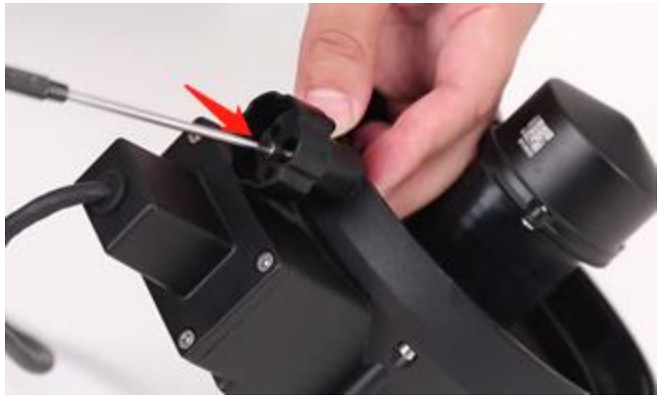
13) Connect the control board with the control cable. Tighten the control cable. Screw the control board with two screws.



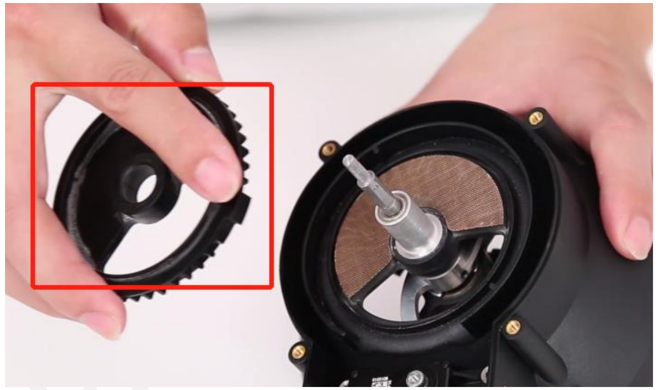
14) Mount the control box's silicon seal.

15) Secure the control box with four screws.

16) Mount the button spring on the spread tank lock knob.

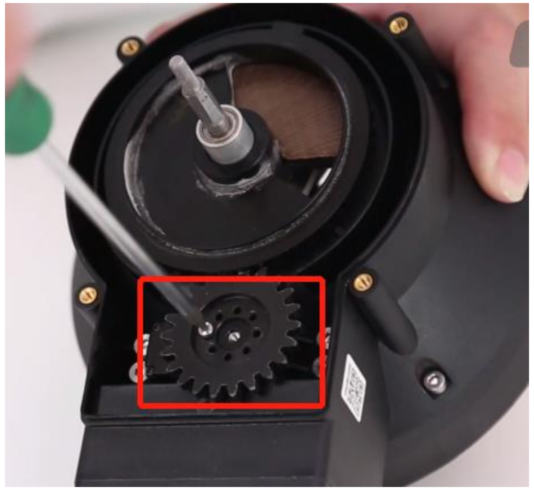


17) Place one screw into the lock knob, align the position and lock the screw.



18) Mount the large gear.

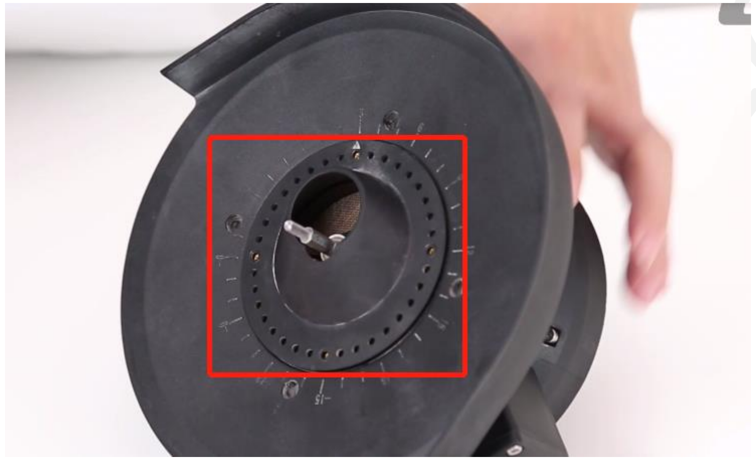
19) Mount the servo disk.



20) Mount the small gear. Rotate it to align the small gear's screw hole with the servo disk. Pre-tighten it with one screw.

21) Rotate the large gear counterclockwise to its limit, and tighten the remaining three screws. Tighten the screws securing the servo disk.

22) Mount the spinner disk's upper plate and screw it with four screws.



23) Mount the Funnel and secure it with four screws (the Funnel's triangular mark aligns with the spinner disk's upper plate scale “3”).

24) Mount the flange, then secure it with two screws.

25) Mount the spinner disk and spinner disk's washer, and secure them with four screws.

26) Mount one M4 anti-skid nut.

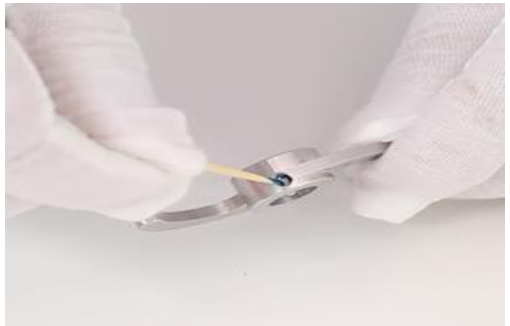
27) Lift up the lock knob. Connect the spreading system module to the spread tank. Rotate them. When the lock knob aligns with the spread tank’s lock hole, tighten the lock knob by 90°.

1) While disassembling the lock knob, be careful not to lose the springs.

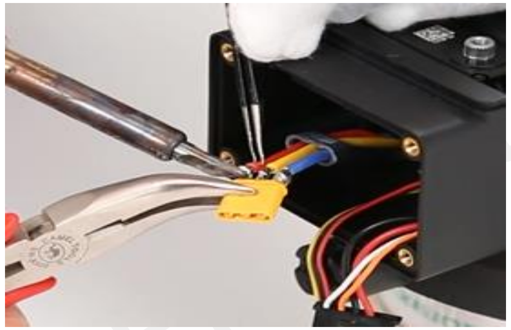
2) While disassembling the control board, push out the cable from behind. Do not drag it.

3) While welding the ESC cable connector, please make sure the cables protected.

4) While cleaning off the white glue on the Hall sensor board, do not damage the surrounding components.



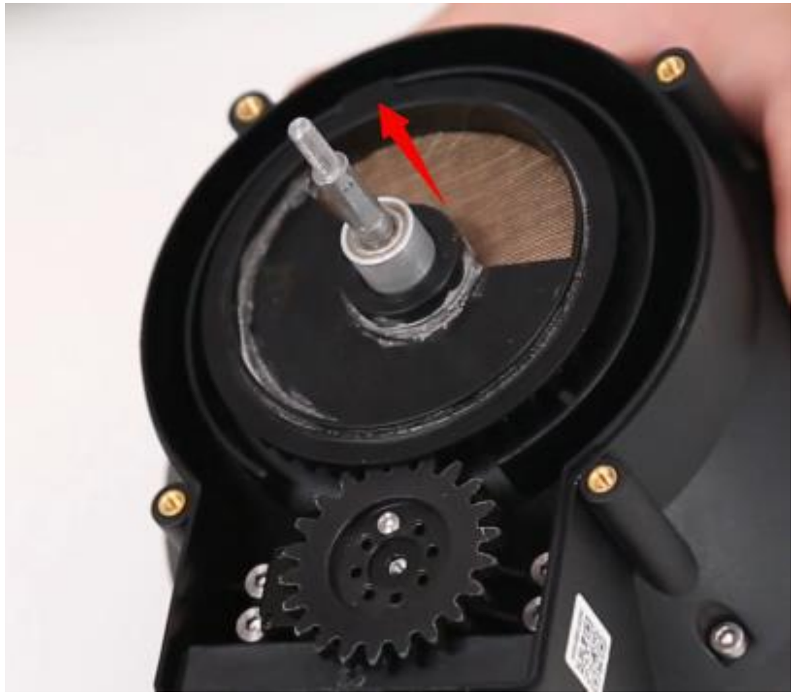
5) In places where screws are used, please apply screw glue on the screw or the screw hole.



6) While welding the ESC cable connector, ensure that the cable sequence is correct. Avoid short circuiting.

7) Before mounting the control board, please arrange the cables in the control box.

8) Be sure to attach the sealing rubber ring in the control box.



9) Before mounting the large and small gears, align the servo.

Servo calibration process:

1. Remove the large gear, and ensure that the large gear does not contact with the servo's small gear.
2. Power on the servo for self-check. After self-check, the servo angle will automatically return to zero.
3. Mount the large gear. Rotate the large gear counterclockwise to its limit (as shown above). Lock the screw that secures the large gear.
4. After mounting, conduct calibration with the app.

03 Damage Assessment

部件	故障	原因分析	处理方案
固件	无法升级	1.交叉检查是否为飞行器端故障	分析飞行器故障原因
		2.检查调参是否为最新版本	官网下载最新版本调参
		3.检查播撒系统控制线是否损坏	更换播撒系统控制线
		4.交叉对换检查控制板是否损坏	更换控制板
	无法连接调参	1.交叉检查是否为飞行器端故障	分析飞行器故障原因
		2.检查电脑是否缺少驱动	更换电脑或重装最新版本调参
		3.检查播撒系统控制线是否损坏	更换播撒系统控制线
		4.交叉对换检查控制板是否损坏	更换控制板

部件	故障	原因分析	处理方案
仓门	仓门无动作	1.交叉检查是否为飞行器端故障	分析飞行器故障原因
		2.检查大齿轮、小齿轮是否有异物卡住	清理异物
		3.检查控制板上舵机线是否松动	确保连接正常
		4.交叉对换检查舵机是否损坏	更换舵机
		5.检查播撒系统控制线是否损坏	更换播撒系统控制线
		6.交叉对换检查控制板是否损坏	更换控制板
	仓门无法完全打开或关闭	1.系统需要重新校准	在 App 内重新校准
		2.检查大齿轮是否被异物卡住	清理异物
		3.检查大齿轮是否损坏	更换大齿轮
		4.检查小齿轮螺丝是否松动	锁紧螺丝
		5.舵机校准不良	拆开大齿轮、小齿轮再进行舵机校准

故障	原因分析	处理方案
甩盘不转	1.交叉检查是否为飞行器端故障	分析飞行器故障原因
	2.检查飞行器、播撒系统固件是否最新	更新固件
	3.检查甩盘主体内部是否有异物卡住	清理异物
	4.检查播撒法兰顶丝是否松动	锁紧顶丝
	5.检查减速箱电调线是否损坏	修复电调线
	6.交叉更换检查电调板是否损坏	更换电调板
	7.检查播撒系统控制线是否损坏	更换播撒系统控制线
	8.交叉对换检查控制板是否损坏	更换控制板
	9.交叉对换检查减速箱组件是否损坏	更换减速箱组件
甩盘转速慢	1.检查甩盘是否有异物卡住	清理异物
	2.检查甩盘主体上的轴承是否卡死	更换甩盘主体组件
	3.交叉更换检查电调板是否损坏	更换电调板
	4.交叉对换检查控制板是否损坏	更换控制板
	5.交叉对换检查减速箱组件是否损坏	更换减速箱组件

部件	故障	原因分析	处理方案
减速箱	甩盘与搅拌棒其中之一不转	1 检查甩盘主体内部是否有异物卡住	清理异物
		2.检查搅拌棒顶丝是否松动	锁紧顶丝
		3.交叉对换检查减速箱组件是否损坏	更换减速箱组件
	甩盘与搅拌棒都不转	1 检查甩盘主体内部是否有异物卡住	清理异物
		2.检查搅拌棒、播撒法兰顶丝是否松动	锁紧顶丝
		3.检查减速箱电调线是否损坏	修复电调线
		4.交叉更换检查电调板是否损坏	更换电调板
		5.检查播撒系统控制线是否损坏	更换播撒系统控制线
		6.交叉对换检查控制板是否损坏	更换控制板
		7.交叉对换检查减速箱组件是否损坏	更换减速箱组件

部件	故障	原因分析	处理方案
霍尔相关	误报无料报警	1.系统需要重新校准	在 App 内重新校准
		2.检查霍尔板线是否松动、破损	确保线材正常
		3.交叉对换检查霍尔板是否损坏	更换霍尔板
		4.检查霍尔支撑的轴承外圈、内圈螺丝胶是否失效	拆下重新打上螺丝胶
		5.检查霍尔支撑件（轴承）是否卡死	更换减速箱组件
		6.交叉对换检查控制板是否损坏	更换控制板
	无料不报警	1.系统需要重新校准	在 App 内重新校准
		2.交叉对换检查霍尔板是否损坏	更换霍尔板
		3.检查霍尔支撑件转动是否顺畅	确保转动顺畅
		4.交叉对换检查控制板是否损坏	更换控制板

04 FAQ

- **Can I adjust the spreading rate of the Spreading System for DJI Agras MG Series drones? How do I adjust it?**

The spreading rate cannot be adjusted directly. You can affect the spreading rate by adjusting other parameters.

Adjust the hopper outlet size and the spinner disk rotating speed to affect the spreading rate. Tap the Hopper Outlet Size on the top-right of the operation page and start editing.

- **What is the compatible particle diameter of the Agras T20 Spreading System?**

The Agras T20 Spreading System is compatible with particles with a diameter between 0.5 - 5 mm.

- **What's the max flow of the Agras T20 Spreading System? What is the spreading range? What's the capacity?**

The Agras T20 Spreading System can spread 15 kg of particles per minute.

The maximum spreading range is 7 m.

The capacity is 20 L.

- **Can I ensure the spreading effect right below the Agras T20?**

The Spreading System allows for 360° horizontal spreading, which can ensure even spreading throughout the flight route.

- **Will the materials spread by the Spreading System 2.0 on the landing gear affect the uniform spreading and the spreading range?**

It will not affect the spreading range. The landing gear covers a very small scope, exerting little influence on the uniformity.

- **What is the maximum loading amount of the Spreading System 2.0?**

When the Agras T20 is used, the maximum capacity is 20 kg. When the Agras T16 is used, the maximum capacity is 15 kg.

- **Is the Spreading System waterproof? Will an inflow of liquid cause a fault?**

The Spreading System is not waterproof. An inflow of liquid will damage the electronic components and corrode the mechanical component.

- **What kinds of materials can be spread by the Spreading System 2.0?**

It can spread rice, rapeseed and other grain seeds, granulated feed for shrimp rice fields, and solid particle fertilizers.

- **What is the spreading range of the Spreading System 2.0?**

The maximum effective spreading range is 7 m. The effective spreading range varies depending on the type of particles spread.

- **Does the Agras T16 support the Agras T20 Spreading System?**

The Agras T20 Spreading System can be mounted on the Agras T16, but the load must be within 15 kg.

- **Where can I connect the Agras T20 Spreading System ?**

The Agras T20 Spreading System is connected to the liquid level meter/flow meter port of the spraying module.

- **How many flights can the Agras T20 battery perform for pesticide spreading?**

The number of flights depends on the amount of material spread per mu.

- **Does omnidirectional obstacle avoidance work while the Aras T20 Spreading System is operating?**

Yes.

■ **How does the spinner disk of the Spreading System work in A-B Route Operation Mode?**

In A-B Route Operation Mode, when Point A is recorded, the spinner disk starts spinning but will not spread. Once the recording is completed, the spreading will begin.

■ **Does the Spreading System support banked turning?**

The Spreading System supports banked turning. However, during banked turning, spreading will not take place and the spinner disk will keep rotating.

■ **Does the Spreading System have pressure value and throttle value in static and dynamic state?**

There is no pressure value or throttle value. It only has two main parameters: hopper outlet size and the rotating speed of the spinner disk, which will not change after setting.

■ **Does the Spreading System have an empty tank warning?**

Yes. An "Empty Spray Tank" warning will appear on the remote controller screen.

■ **Calculation of spray amount per mu for Agras drones**

Calculation of spray amount per mu

Specify the amount of pesticides used. The recommended amount to spray is 1 kg/mu.

Planned flight speed (varies depending on efficiency. The recommended value is 4 m/s).

Adjust the rotating speed of the spinner disk (Measurement is required. The recommended spreading range is 4 - 6 m).

Spray amount per mu/(666/spreading range/speed/60) = Material delivery weight per minute. Adjust the hopper outlet size.

Example:

Supposing that the required spray amount is 0.55 kg/mu, the planned flight speed is 4 m/s, and the spreading range is 5 m, how do I adjust the hopper outlet size?

Based on 0.55/(666/5/4/60)≈1, adjust the hopper outlet size to ensure that the material delivery weight per minute is 1 kg. Static testing can be performed.

Thank you